DunesForge Quickstart & Handbook

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Everything lives in one place: press **[N]** in the 3D viewport and pick the **TyDunes** tab.

> **A note on the name.** The product is **DunesForge**. Inside Blender the add-on's panel is still labelled **TyDunes** - that is the tab to look for under **[N]**, and the file you install is `tydunes_blender.zip`. The interface will catch up with the name; nothing about how it works changes.

How to read this document

You do not need to read it front to back.

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Impatient	§1 Five minutes - open a finished scene, press Play
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1. Five minutes to your first dunes

Do not start from a blank scene. Start from a finished one and take it apart - it is faster and you will learn more.

Open a demo

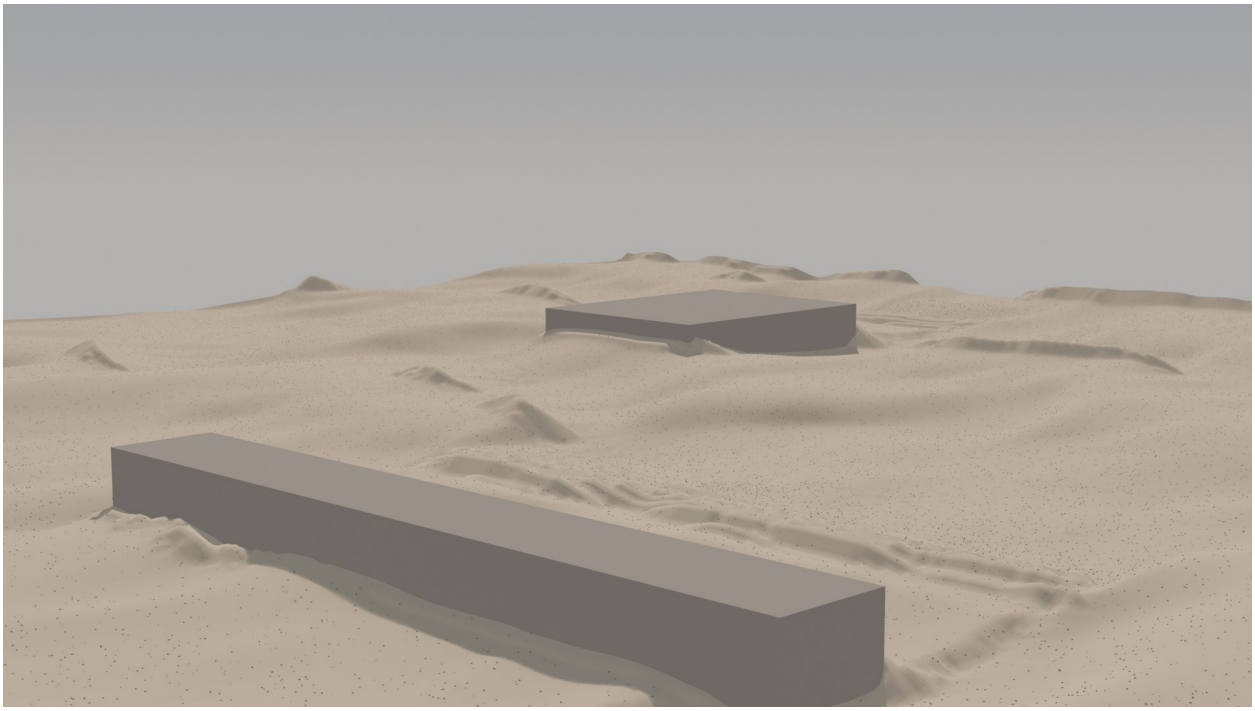
1. New Blender file. (You can delete the default cube; DunesForge ignores it.)

2. [N] > **TyDunes** tab.

3. Open **Presets > Demo Scenes**, choose **Desert: Dunes + Obstacles**.

4. Press **Create Demo Scene**.

Wait a few seconds - the scene runs a warm-up so it opens with dunes already formed rather than with flat ground. You get terrain, two rock blocks with sand tails behind them, a camera, lights, a material, and a text card in the corner explaining what the scene demonstrates.



Dunes with two rock blocks, wind shadows and lee tails

5. Press **[Spacebar]** to play. The dunes migrate downwind.

6. While it plays, grab one of the rock blocks and move it. **The wind shadow

follows it**, and sand begins piling up on its new upwind side.

That last step is the product in one gesture: you did not sculpt the drift, you moved an object and the drift followed.

> **Tip:** the text card is an ordinary Blender object called `TyDunes_Demo_Label`, plus its backing plate `TyDunes_Demo_LabelBack`. Hide both when you want a clean render.

Now build one from nothing

Reading a finished scene teaches you what is possible; building one teaches you what each step does.

1. Reset Simulation - clears any previous solver state.

2. Initial Heights > `Flat`.

3. Initial Sand Depth > `1.0`. *(A flat sheet of loose sand.)*

4. Initialize Simulation. The mesh appears - flat, and that is correct.

5. Wind Direction > `0`. **Wind Strength** > `1.0`.

6. Press **Run Steps**. It simulates 100 steps with no viewport redraw, which

is far faster than playback.

7. Press it four or five more times.

What you are watching: flat sand is unstable under wind. Tiny irregularities grow because a bump slows the wind behind it, sand drops there, and the bump gets bigger. That runaway is why dunes exist, and you did not have to model it - it falls out of the transport equations.

Try this immediately: set **Wind Direction** to `90` and run more steps. The whole field reorganises to face the new wind. This is the thing that hand-built dune fields cannot do.

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2. Don't panic: the interface has three levels

DunesForge exposes a great many controls, because a great many people needed different things from it. **You are not expected to look at all of them.**

At the top of the panel there is an **Interface** selector:

Level	What you see	Who it is for
Simple *(default)*	Demos, terrain, wind, basic weather, appearance	Everyone, on day one
Standard	The working set: physics, objects and masks, water, plants, multi-fraction grain, extra phenomena (tides, snow layers, slab avalanches, playa), surface detail, export	Anyone building a real shot
Advanced	Everything, including the pressure solver, planetary constants, tracers and experimental controls	TDs, and the curious

Three things make this safe to play with:

and the solver keeps reading it.

the same scene, number for number.

never be quietly doing something you have no way of seeing.

- **Nothing is ever disabled - only hidden.** A hidden control keeps its value,
- **Switching levels cannot change your simulation.** Simple and back gives you
- **Hidden-but-active systems are listed near the selector**, so a scene can

Practical advice: stay in **Simple** until something you want is not there. Then go to **Standard**. Reach **Advanced** when you know what you are looking for - that level is a toolbox, not a tutorial.

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3. How DunesForge actually works

Five ideas. Once you have them, nothing in the add-on is mysterious.

3.1 There is a simulation, and there is a mesh

The solver keeps a **height field** - a grid of heights, like a heightmap. The Blender mesh is a *picture* of that grid, rewritten every time the simulation advances.

Consequences worth knowing:

sets how finely the physics is computed. `Render Subdivision` controls how much geometry you get out.

overwrites it. Use the **Interactive Brush** instead - it paints the solver's own fields.

hand it to someone without the add-on.

- **Grid resolution is not mesh resolution.** `Resolution` (in the main panel)
- **Sculpting the mesh by hand does nothing to the simulation.** The next step
- **The mesh is an ordinary Blender object.** Render it, shade it, cache it,

3.2 Everything physical is off by default, and does nothing at its default

Snow, water, rain, plants, frost, the avalanche extras, the airborne systems - all opt-in. A scene you have not touched runs exactly the arithmetic it ran before those features existed.

This is not a slogan; it is why you can update the add-on without your saved shots changing.

3.3 Two clocks: Run Steps and Play

This one catches everybody once.

	Run Steps	Play (timeline)
What it does	Advances the solver directly, in a batch	Advances frames
Viewport	Not redrawn until it finishes	Redrawn every frame
Speed	Much faster per step	Slower
Reads animation keyframes	**No**	**Yes**
Advances the solver	**Yes**	**Yes**

Run Steps advances the physics, and that includes the shallow-water field. Measured on *Pools That Flow*: a single batch of 40 solver steps advanced the water by 1205 of its own steps and moved the surface by up to 0.344 m, with the total volume unchanged to six decimal places.

What Play adds is the timeline. Animation keyframes are read by frame playback and by nothing else, so anything you keyframed - a moving obstacle, a gate that lifts, a wind direction that swings on keys - needs Play. Scene dressing that refreshes per frame also looks right only in playback.

So: if you keyframed something and it is not moving, you are probably pressing Run Steps when you want Play.

3.4 The wind is a field, not a direction

`Wind Direction` is the *base* heading. What the sand actually feels is a shear field computed across the whole terrain, which speeds up over crests and separates behind them, leaving a sheltered pocket on the lee side.

That is why:

- obstacles cast **wind shadows** that reach downwind of themselves,
- crests are **scoured** while troughs **fill**,
- a canyon **channels** and accelerates the flow through its narrows.

None of this is painted on. It comes from the wind model, and it is the reason the results respond correctly when you move things.

3.5 Sand has a maximum slope, and that is what sells the look

Loose grains cannot hold a slope steeper than the **angle of repose** - about 34 deg for dry sand. When deposition pushes a face past that angle, it **avalanches** until it is back at the limit.

Every dune in your shot gets the same lee-face angle, because it is set by physics rather than by taste. Viewers do not know the number, but they notice when it is inconsistent.

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4. A guided tour of the panels

What each panel is for, and when you would open it. Full control-by-control detail is in **REFERENCE.md**.

Panel	What it is for	You open it when...
DunesForge (main)	Terrain, resolution, scale, the simulation buttons	Always - this is the simulation
Presets	Complete starting points, and the 66 demo scenes	Starting anything
Wind	Direction, strength, variation over time, wind models, temperature	You want the wind to do something specific
Objects / Masks	The slots that let scene objects talk to the sand	Your shot contains anything that is not terrain
Physics	Repose angle, avalanching, separation, dune length scale	The material should behave differently
Water / Moisture	Water table, flowing water, rain, wetness	Coasts, oases, rivers, weather
Snow	Switches the material model to snow; sintering, slabs	Winter, and all avalanche work
Vegetation	Plants that grow, shelter sand, get buried, die	Semi-arid ground, nabkha mounds
Interactive Brush	Painting sand, wind, masks, tracks by hand	Art direction; when physics needs persuading
Surface Shader	How texture is projected onto moving geometry	The camera gets close enough to see surface
Surface Detail	Ripples and micro-relief as geometry	Close-ups

Panel	What it is for	You open it when...
Sand Haze	Airborne sand lifting off the crests	You want the wind to be felt in a still frame
Dust Devils	Travelling vortices that pick up and drop material	Desert middle distance, something alive
Crest Backlight	A light ribbon along the crest line	Sunrise/sunset framing
Multi-Fraction	Two grain sizes, sorting, pavement, megaripples	Ground that needs a history
Field Maps	Load/save the solver's fields as images	Round-tripping, reusing painted masks
Export / Bake	Meshes, caches, displacement and material maps	Handoff, and when you want the cost to stop
Performance	The speed/quality trade-off	Too slow, or fast enough and you want more
Extra Phenomena	Frost, ice, thermal effects, specialised processes	Specific shots that need them
Debug / State	What the solver currently holds, in numbers	Something is not doing what you expect

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5. A working method, from idea to render

A sequence that avoids the usual dead ends. Steps 1-4 are cheap; do not skip them to get to the pretty part faster.

Step 1 - Decide what kind of ground this is

Before touching anything, answer two questions:

1. Is there a lot of sand, or a little? A thin veneer over rock gives

separated crescent dunes that migrate. A deep bed gives connected ridges. This is `Sand Supply` and `Initial Sand Depth`, and it changes the shot more than any other choice.

2. Is the terrain flat, or does it have relief? `Initial Heights` gives you

procedural options; `Terrain` lets you point at your own object.

Step 2 - Work small and fast

Set **Resolution** low while you experiment. You are looking for the right *behaviour*, not the final detail, and a coarse grid gets you there in a fraction of the time. Raise it when the behaviour is right.

Step 3 - Get the dune scale right early

If the dunes come out the wrong size, change **Dune Scale** - not your terrain dimensions. The solver has its own length scale, and Dune Scale is the knob that maps it onto your world. Fighting this with terrain size will cost you hours.

Step 4 - Add the interactions, one at a time

Obstacles, sources, water, plants. **One at a time, running steps after each.** If you turn on five systems and the result is wrong, you have five suspects and no evidence.

Step 5 - Run it out

Run Steps repeatedly until the field has developed. Dunes take time - a freshly initialised scene is not a dune field, it is the beginning of one.

Step 6 - Detail and dressing last

Surface detail, shader, haze, dust devils, crest backlight. These are appearance; doing them before the shapes are right means redoing them.

Step 7 - Freeze, bake, hand off

When the simulation is where you want it, stop paying for it: commit the mesh, bake maps, or cache. See Tutorial 14.

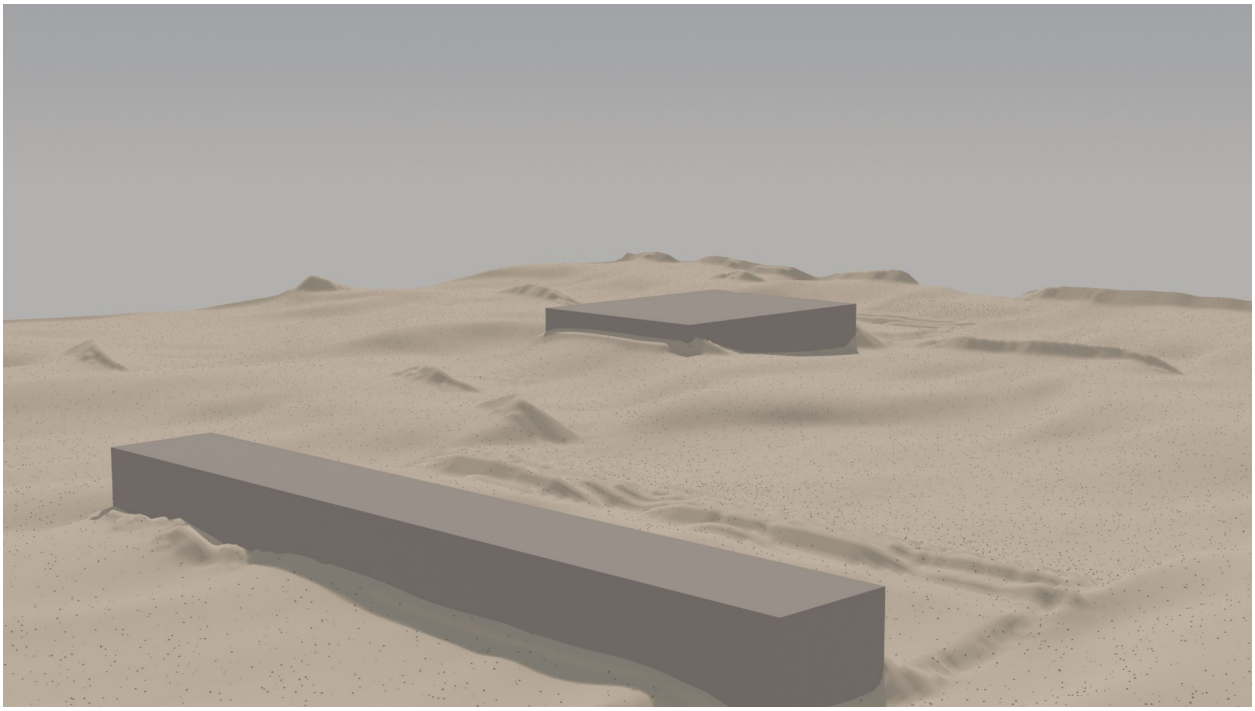
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6. Tutorials

Sixteen recipes. Each assumes a new Blender file with the TyDunes tab open, unless it says otherwise.

Tutorial 1 - A desert that looks like a desert

Goal: a wide sand sea with crescent dunes, not a noise bump map. **Time:** 10 minutes.



Dunes with two rock blocks, wind shadows and lee tails

1. Reset Simulation.

2. Initial Heights > `Flat`; **Initial Sand Depth** > `1.2`.

3. Sand Supply > `0.35`.

4. Wind Direction > `0`; **Wind Strength** > `1.0`.

5. Initialize Simulation, then **Run Steps** four or five times.

Why supply 0.35? Low supply is what makes *separated* dunes. With plenty of sand everywhere, neighbouring dunes merge into continuous ridges. With a little, each dune is an island on bare ground and takes the crescent form.

Look at the crests. Every lee face sits at the same angle. That consistency is the angle of repose, and it is most of why this reads as real.

Variations worth trying:

Change	Result
Sand Supply > `0.8`	Connected transverse ridges instead of crescents
Wind Strength > `0.5`	Everything develops more slowly, finer forms
Dune Scale up	Fewer, larger dunes across the same terrain
Angle of Repose > `28`	Looser material; lower, wider forms
Seed changed	A different field with the same character

Tutorial 2 - Sand that respects your set

Goal: a building, rock or vehicle that sand actually piles against. **Time:** 10 minutes. **Prerequisite:** Tutorial 1, or the *Desert* demo.

1. Place your object on the terrain, sitting on the surface.

2. Objects / Masks > set it as **Obstacle**, and turn on **Use Obstacle**.

3. Run Steps a few times.

You get a drift on the upwind side, scour where the wind accelerates around the corners, and a tail reaching downwind.

Then press Play and move the object. The shadow travels with it.

Controls that matter here:

step into the simulation that it then has to live with.

touching it. Raise it if your object floats slightly.

- **Obstacle Smoothing** - softens the footprint edge. A hard-edged mask puts a
- **Obstacle Collection** - use a whole collection instead of one object.
- **Echo Strength / Echo Distance** - a secondary drift further downwind.
- **Max Float Gap** - how far above the ground an object still counts as

Other object slots, and what they are for (each is a genuinely different job, which is why they are separate):

Slot	What the object does
Obstacle	Blocks wind and transport - the wind shadow
Bedrock	Holds the ground down; sand collects on it but it does not erode
Source	Feeds sand into the scene from its footprint
Snow Source	The same, for snow
Track Object	Presses its *shape* into the surface - see Tutorial 13
Wind Object	Takes over the wind heading from its orientation
Water Bodies	Ponds at their own heights, above the single Water Level

> **The difference between Obstacle and Bedrock** confuses people. An obstacle is a *thing in the wind*. Bedrock is *ground that will not blow away*. A rock outcrop is usually both.

See it done: the **Showcase: Object Masks** and **Showcase: What Objects Tell The Sand** demo scenes put several of these slots in one frame.

Tutorial 3 - Choosing your dune type

Goal: get barchans, transverse ridges or star dunes *on purpose*. **Time:** 20 minutes.

Dune form is not a preset - it comes out of three things: how much sand there is, how variable the wind is, and how long it has run.

Barchans - crescents with horns, on bare ground

- **Sand Supply** low (`0.2`-`0.35`), **Initial Sand Depth** low (`0`-`0.5`)
- **Wind Direction** fixed, **Wind Wobble** low (`0.05`)
- Run a long time. Barchans need to separate from each other.

Transverse ridges - long ridges across the wind

- **Sand Supply** high (`0.7`-`1.0`), **Initial Sand Depth** `1.0`+
- **Wind Direction** fixed, low wobble
- These form faster than barchans.

Star dunes - several arms, no single downwind direction

Star dunes need a wind that changes direction, which is what **Wind Mode** is for. Look at `NORMAL` versus the other modes, and at **Phase Steps** - how long the wind holds each heading before moving to the next. A wind that swings and holds builds arms; a wind that jitters just blurs the field.

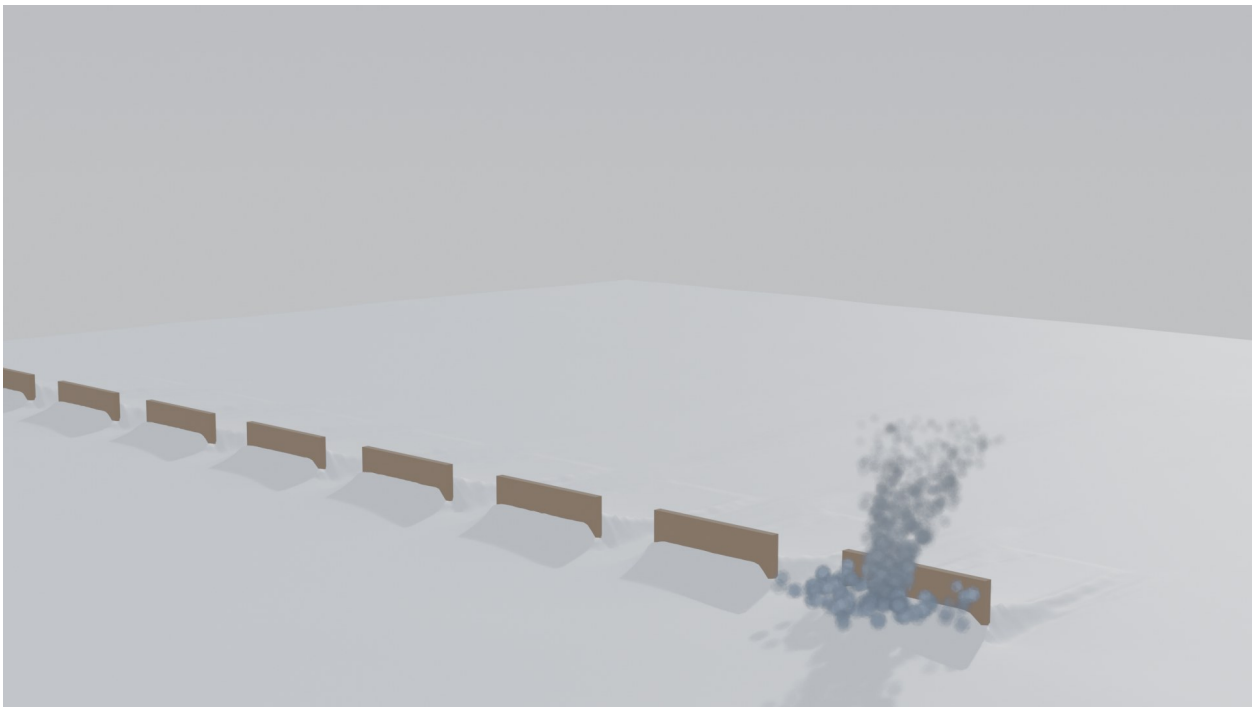
Making any of them less regular

Control	Effect
Wind Wobble	Slow drift of the heading - organic meander
Wind Jitter	Fast randomness - roughens crest lines
Gust Strength + **Gust Hold Min/Max**	Episodic strong wind
Turn Time (steps)	How quickly the wind swings to a new heading

> **A caution:** real dune fields are more regular than people expect. Adding lots of randomness usually reads as *noise*, not as *nature*.

Tutorial 4 - Snow drifts and a fence

Goal: winter, not white sand. **Time:** 15 minutes.



Snow drifting against a slatted fence

1. Open the **Snow: Drifts + Fence** demo, or start from a snow **Preset**.

2. The essential switch is **Snow** in the Snow panel.

3. Put a fence or wall across the wind and make it an **Obstacle**.

4. Run Steps.

Snow piles into the classic wedge on the lee side of the fence.

What makes snow *snow* rather than white sand:

Control	What it models
Sinter	Snow hardens with age. Old drifts resist the wind while fresh powder keeps moving - this is the single most important difference
Initial Hardness	How set the existing snowpack is at the start
Snowfall	New snow arriving - fresh powder on top of hardened base
Melt	Snow leaving
Snow Repose Offset	Snow holds a steeper face than sand
Full Cover Depth	How deep before the ground reads as covered

Try this to see sintering: run the scene with **Sinter** at `0`, then again at its preset value. Without it, every drift is equally erodible and the field keeps rearranging. With it, drifts set and persist.

Tutorial 5 - An avalanche you can stage

Goal: a slab that releases and runs downhill. **Time:** 20 minutes.



A snow slab that released and ran downhill

1. Open the **Snow: Slab Avalanche** demo. This is one case where starting from

the demo is strongly advised: an avalanche needs a slope, a layered pack and a trigger, and the demo sets all three correctly.

2. Press Play and watch the release.

3. Now look at what the scene turned on: a snow preset carrying the layer

stack, a sculpted slope, and the trigger settings.

The machinery, in order:

1. Snow layers build up from snowfall - a stack with different hardness.

2. Slab Weakness Drop creates a weak layer beneath a harder slab.

3. Slab Trigger decides when it lets go.

4. Episodic Slabs makes releases happen as discrete events rather than

continuously.

5. Angle of Repose and **Avalanche Passes** govern how the debris runs out.

Related demos:

growing on the way down.

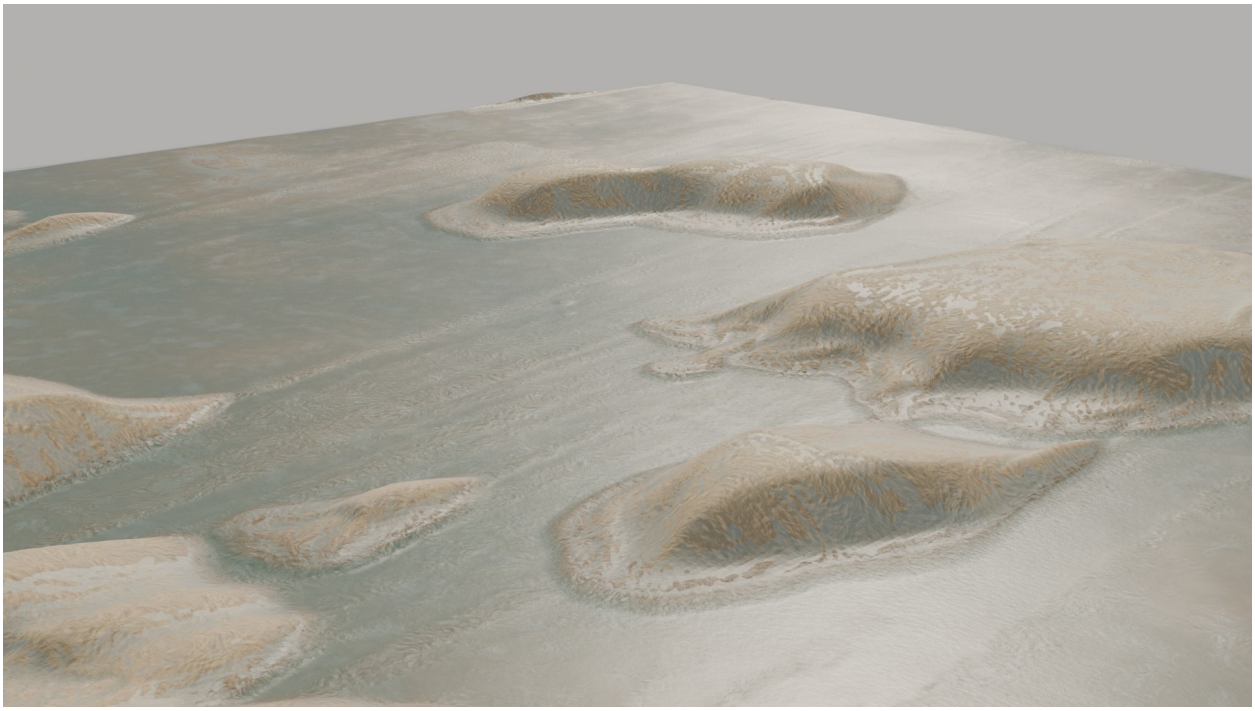
gravity relaxation with no wind at all, which is the cleanest way to see what the repose angle does.

- **Avalanche That Scours Its Path** - the slide entrains material as it goes,
- **Avalanche Into A Valley** - run-out onto flat ground.
- **Slab Avalanches Held Back By Gates** - barriers that catch it.
- **Sand: Slide Under Gravity** / **Snow: Sluffs Off A Rock Face** - pure

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Tutorial 6 - Water that shapes the ground

Goal: a stream that cuts its own channel. **Time:** 20 minutes.



Water running downhill, wetting the sand it flows over

1. Open **Water: Pools That Flow**.

2. Press **Run Steps**, or Play - both advance the water. Run Steps gets you

somewhere faster; Play is what you want once the scene contains keyframed objects such as a sluice gate.

Two things happen at once: water runs downhill as a shallow-water field, and it **wets the sand underneath**. Wet sand resists erosion, so the channel undercuts its own banks instead of spreading into mush.

The three water systems, and which one you want:

System	What it is	Use it for
Water / Moisture (water table)	A *level*. Everything below it is submerged and cannot blow away	Coastlines, oases, lakes at rest
Flowing Water (2D field)	A real shallow-water simulation that runs downhill and can carve	Rivers, floods, sluices, run-off
Rain	Surface moisture - weather that wets ground without simulating drops	Weather, vegetation, damp sand

Doing it yourself:

1. Turn on **Water / Moisture**, set a **Water Level**.

2. Turn on the flowing-water field.

3. Give the water somewhere to go. The domain edges are periodic by

default, so water wraps around instead of leaving. The **Open X / Open Y** edge options let it drain out.

4. Keep an eye on the water field's own resolution and time step - it is

separate from the dune grid and it is where the cost lives.

> **Watch the cost.** The demo ships at a setting that runs smoothly. Raising the water resolution can get expensive very quickly, especially on steep terrain where the solver has to take small time steps.

Related demos: *Reservoir Behind A Sluice Gate* (a gate that lifts and releases a flood), *Ripples Carried By The Current*, *Wind On The Lake*, *Ice: Pack, Ridges and Leads*.

Tutorial 7 - Rain, damp sand and drying out

Goal: ground that has weather on it. **Time:** 15 minutes.

1. Open the **Rain: Wet Sand + Plants** demo, or turn on **Rain** yourself.

2. Press Play.

Rain is not a water simulation - it is a surface-moisture model, and that is the right tool for weather. Rain raises moisture; moisture makes sand harder to lift; sun and wind dry it out again.

Why you want it even in a dry shot: damp sand does not erode. A shower leaves a crust that the wind has to work through, so the surface gets a history rather than looking uniformly blown.

Controls to know:

wet hollows and dry crests, which is what makes vegetation and species behaviour interesting.

- **Coverage** - whether rain falls everywhere or in patches. Below 1.0 you get
- **Infiltration / Hold** - how fast water soaks in and how long it stays.
- **Drying (wind and sun)** - the return to dry.
- **Crust** - a hardened skin left after the water has gone.

Combining with temperature: the Wind panel has a temperature section - air temperature, a sleet band, melt per degree, rain-on-snow. With it you can make the same weather fall as rain or as snow depending on how cold it is.

Related demos: *Rain: Wet Sand + Plants*, *Monsoon*, *Oasis*, *Drying*, *Rain On Water*, *Frost: Cold Dawn*, *Frost: Hoar on Snow*.

Tutorial 8 - Plants, and the mounds that form around them

Goal: semi-arid ground that looks inhabited, and nabkha dunes. **Time:** 20 minutes.



Two plant species separating across the terrain by how well they tolerate drought

1. Turn on **Vegetation**.

2. Set **Initial Cover** to something small, like `0.05`.

3. Growth controls how fast plants spread; **Burial** how easily sand kills

them; **Wind Shield** how much they protect the ground they stand on.

4. Run Steps repeatedly.

The interesting part is the feedback loop. Plants grow where sand is calm. Plants shelter sand. Sheltered sand is calmer, so more plants grow. Eventually a storm buries the lot and it starts again.

Nabkha - the mound that forms around a bush:

Raise **Nabkha**. Each plant traps sand around itself and sits on a growing mound. This is a real and very recognisable desert landform, and it is one of the clearest demonstrations that the vegetation is coupled to the sand rather than scattered on top of it.

Blowout is the opposite: where vegetation fails, the wind opens a hollow and eats into the field.

Where plants may and may not grow:

Control	What it limits
Min / Max Slope	Keeps plants off faces that are too steep
Min / Max Height	Keeps them to a height band - a shoreline, a valley floor
Ragged Limit Edge	Stops the limit reading as a drawn line
Limits Apply to Seeding	Whether limits govern where seeds land as well as where plants survive
Keep Off Sand Source	Plants avoid where sand is being fed in

Two species: Second Species plus **Tolerance (Species B)** gives you two plants with different drought tolerance, which separate across the terrain wherever moisture varies. It needs *uneven* moisture to mean anything - with uniform wetness both species see the same conditions everywhere.

Life and death: Lifespan, **Senescence**, **Age Slows Growth**, **Dieback**, **Drought Dieback** - a stand that establishes, matures and fails rather than growing forever.

Related demos: *Plants: Two Species Side By Side*, and the vegetation demos covering dieback, nabkha and seeding.

Tutorial 9 - Surface detail worth rendering close up

Goal: ripples and texture that hold up when the camera comes in. **Time:** 15 minutes.

1. Build any dune field.

2. Surface Detail > turn on **Show Render Detail**.

3. Ripple Amp and **Ripple Length** set the size of the ripples.

4. For a one-off pass of fine relief on the current surface, press

Add Micro-Detail Pass.

Why this is not a normal map: these ripples are *geometry driven by the wind field*. They wrap around the dune and change direction where the wind does. A tiling normal map cannot do that, and the difference is obvious in a slow camera move.

Controls worth understanding:

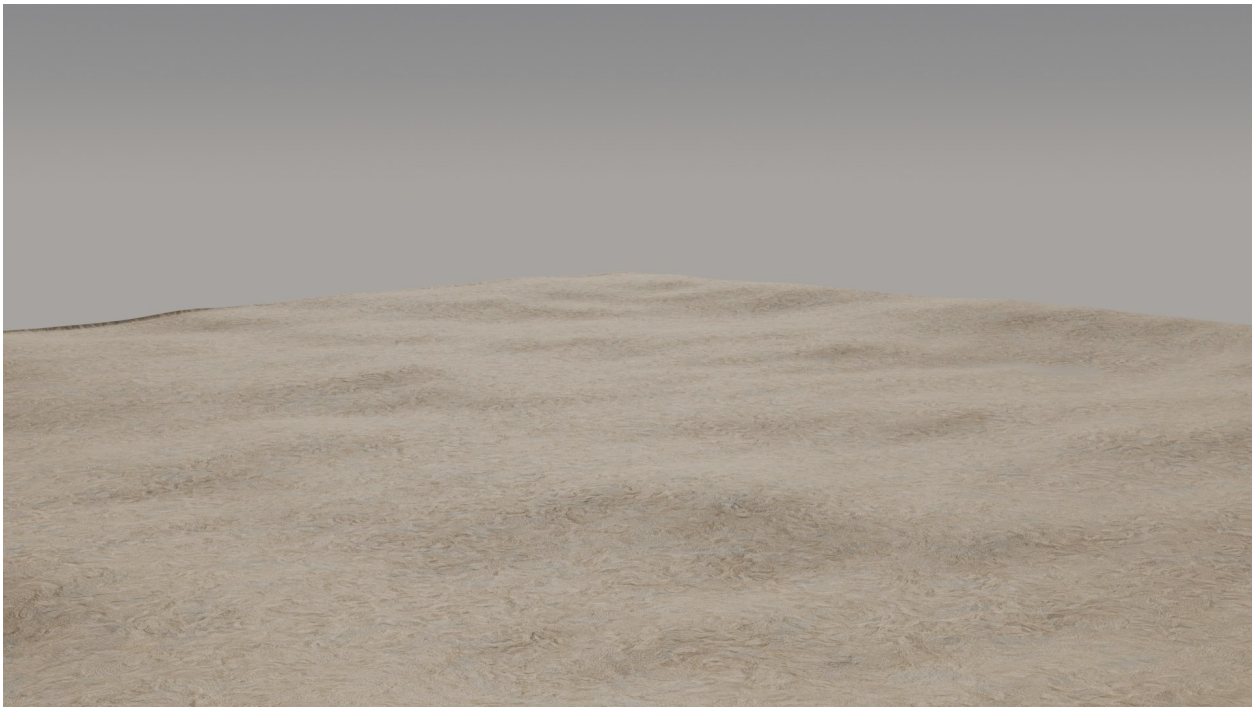
Control	What it does
Ripple Style	Dunelets vs carved vs other styles - different physical character

Control	What it does
Ripple Length Follows Grain	Ripple wavelength scales with grain size, as it does in reality
Ripple Warp	How much the wind field bends the ripple lines
Slope Fade	Fades ripples on steep faces, where they do not survive
Sharpness / Polish	Crest definition versus smoothness
Detail Feeds Sim	Whether the detail layer affects the simulation, or is render-only
Physical Ripple Solver	A real ripple simulation rather than a procedural layer

See it done: Showcase: Surface Detail is a scene built with a close camera for exactly this.

Tutorial 10 - Grain sorting: pavement and megaripples

Goal: ground with a history - coarse gravel armouring, megaripples. **Time:** 20 minutes.



Grain-size sorting: coarse pavement in the troughs, fine sand on the crests

Real sand is not one grain size, and what happens when the wind sorts it is visually distinctive.

1. Turn on **Multi-Fraction Grain**.

2. Coarse Fraction sets how much of the bed is coarse.

3. Winnowing is the wind removing fines and leaving coarse behind.

4. Lag Armouring is the coarse layer then protecting what is underneath.

5. Run Steps for a long time - this is a slow process.

What you get: desert pavement. Areas that have been blown for a long time lose their fines and end up armoured with coarse grains, which then resist further erosion. It reads as *age*.

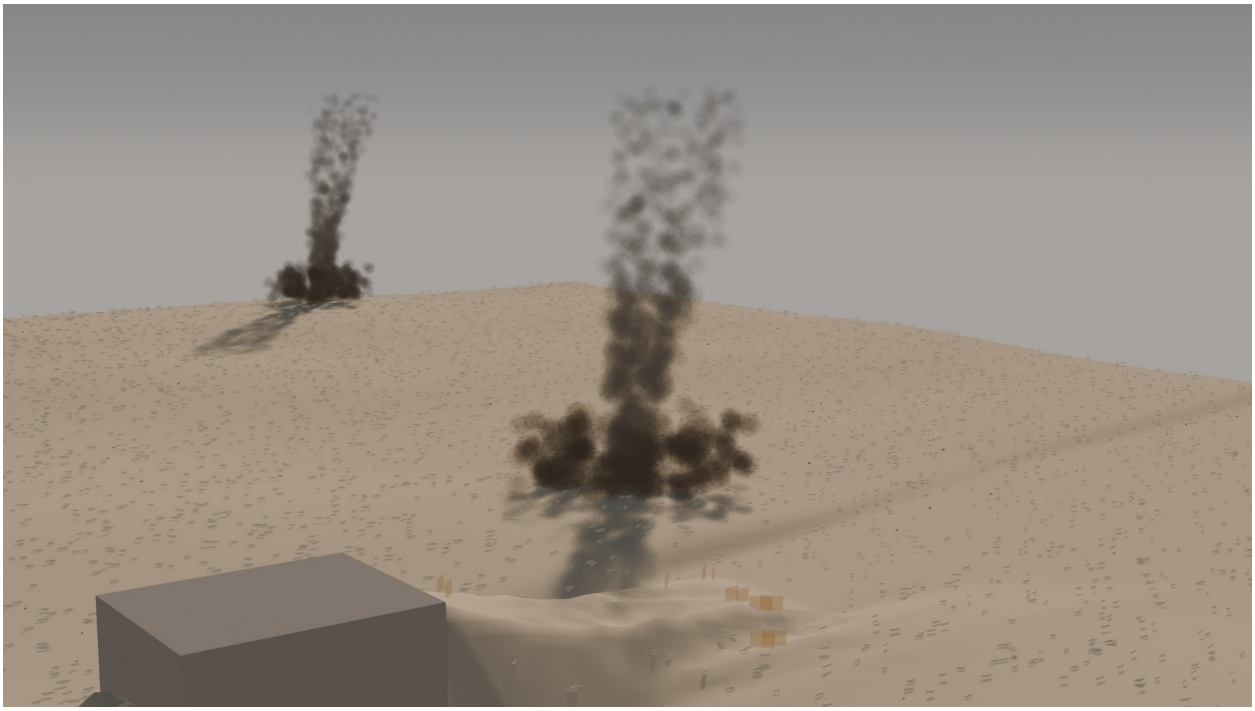
Megaripples are the same mechanism at a larger scale: **Megaripple Height** gives you the large coarse-crested ripples that form where two grain sizes are present.

The simpler two-grain system: Transport Two Grain Sizes with **Coarse Share at Start**, **Coarse Grain Size** and **Coarse Rolls Further** is a lighter-weight version of the same idea.

Related demo: Showcase: Grain Systems puts the grain-scale models side by side in one frame.

Tutorial 11 - Atmosphere: haze, dust devils, backlit crests

Goal: wind you can see, and something alive in the middle distance. **Time:** 20 minutes.



Airborne sand, dust devils and dressing, over dunes that can be frozen in place

These are the systems that sell *air*. All are frame-playback features - press Play, not Run Steps.

Sand Haze - the sheet lifting off the crests

Turn on **Sand Haze**. Grains lift from crests and drift downwind. In a still frame, this is what tells the audience there is wind.

Key controls: **Max Grains** (cost), **Haze Render** (points or instances), **Particle Size / Stretch**, **Custom Particle** if you want your own geometry, and **Haze Update Every** to trade smoothness for speed.

Dust Devils - travelling vortices

Turn on **Dust Devils**. **Count** sets how many, **Size / Radius** and **Column Height** their scale, **Drift Speed** and **Wander** how they travel.

The interesting part is ground coupling. A devil can read the surface under it:

Control	What it makes the devil do
Needs Loose Sand	Weaken over ground with nothing to pick up
Needs Moving Sand	Only thrive where sand is already in motion
Damped by Wet and Plants	Die over damp or vegetated ground
Needs Fresh Snow	Draw from fresh snow instead of sand - snow devils
Respawn On Loose Ground	New devils appear where there is material

Colour controls (**Base / Tip / Skirt Colour**, **Variation**) let you match them to your ground.

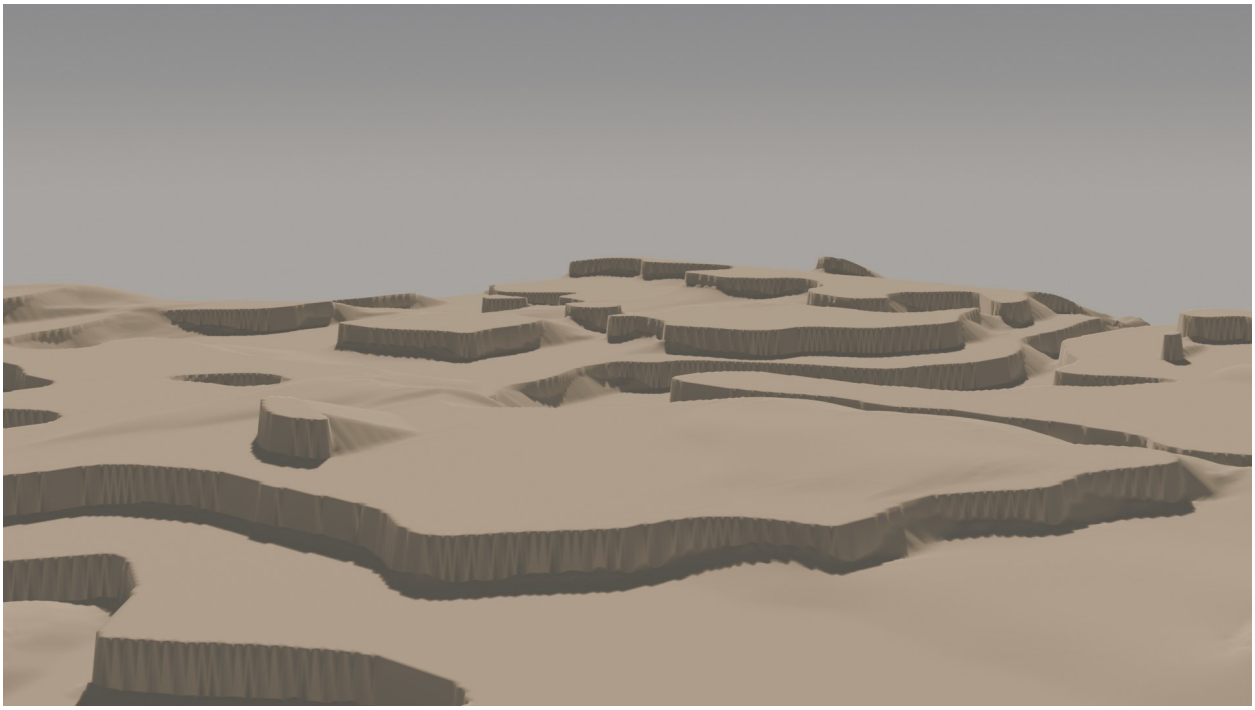
Crest Backlight - the ribbon along the crest

Crest Backlight with **Crest Sheet Height** and **Crest Backlight Tint** gives a thin band of light along the crest line. For sunrise and sunset framing, where the crest line *is* the composition.

See them together: Showcase: Airborne + Dressing is built for this, and it demonstrates the freeze switches - dunes hold while the air keeps moving.

Tutorial 12 - Starting from your own terrain

Goal: simulate on a landscape you made or imported. **Time:** 15 minutes.



A layered mesa: sand settles on the shelves and in the hollows, the steep walls stay bare

There are three routes in.

Route 1 - Point at a Blender object

Put your mesh in the **Terrain** slot in the main panel. It becomes the simulation surface.

Route 2 - Load a heightmap image

Initial Heights > the image option, then set **Heightmap File**. Controls you will want:

smooths them into a continuous surface.

needles.

- **Image Amplitude** - how tall the heightmap reads.
- **Image Offset** - shifts the whole thing vertically.
- **De-band 8-bit Heightmap** - 8-bit images arrive in visible steps; this
- **Remove Spikes** - kills single-pixel outliers that would otherwise become

Route 3 - Generate one

Initial Heights has procedural options: ridged, turbulence, FBM, dune ridges, craters and more, with amplitude, scale, octaves and warping.

The decision that matters: rock or sand?

Start Terrain = Rock treats the whole starting terrain as bedrock - the wind will not erode it, and sand collects on top. Leave it off and the terrain is loose material the wind can carve away.

There is also **Terrain = Sand**, which is the opposite intent: treat an imported or sculpted landscape as loose sand, so its hills deflate, migrate and can vanish entirely. The **Terrain: Dune Field** demo uses this - a dune field used as starting relief, which then behaves like actual dunes.

Tutorial 13 - Tracks, footprints and hand-painted sand

Goal: vehicle ruts, footprints, and art-directed sand. **Time:** 20 minutes.

Tracks along a curve

In the Interactive Brush panel: set a **Track Curve**, pick a **Track Style** (rut or otherwise), then **Track Width**, **Track Depth** and **Berm Height**. You get a vehicle track that follows your curve, with material pushed up at the edges.

Footprints

Print Shape (boot and others), **Print Length**, **Stride** and **Track Width** give a line of footprints with a believable gait.

Objects that press their own shape in

Different from an obstacle: **Objects Leave Tracks** stamps what actually *touches* the ground. A log lying on sand leaves a log-shaped impression, not a rectangle.

Controls: **Contact Gap** (how close counts as touching), **Edge Softness**, **Depth in Sand / Depth in Snow**, **Rim in Sand / Rim in Snow** (material pushed up around the edge), and **Imprint Every Frame** for moving objects.

> **This is a frame-playback feature.** Imprints happen on Play. If you press Run Steps and see nothing, that is why.

Painting by hand

The brush paints the solver's own fields, so the simulation carries on from what you painted:

brush.

- **Paint** - which field: sand, wind, masks, and others.
- **Mode** - add, subtract, set to a target value.
- **Radius / Strength / Falloff** - the usual brush controls.
- **Stamp** and **Stamp Image** - stamp a shape or an image instead of a round
- **Scatter Count / Scatter Clump Size** - scatter rather than a solid stroke.
- **Symmetry** - mirror your strokes.
- **Show Painted Mask** - see what you have painted as an overlay.

When to reach for this: when the shot needs a dune in a specific place and the physics will not be talked into putting one there. Paint it, then run steps - the simulation will make it *behave* correctly even though you placed it.

Tutorial 14 - Getting it out: meshes, maps and caches

Goal: a result you can render, hand over, or stop paying for. **Time:** 15 minutes.

Commit the mesh

When the simulation is where you want it, commit it. You get an ordinary Blender mesh; the solver stops costing you anything.

Commit With Render Detail decides whether the surface-detail layer is baked into that geometry or left as a render-time effect.

Bake displacement and material maps

For a shading pipeline, bake maps instead of carrying dense geometry:

one automatically.

- **Displacement Map Size** - resolution of the baked map.
- **Map Format** - EXR by default, which is what you want for displacement.
- **Render Subdivisions** - how finely the displacement is applied at render.
- **Hide Sim Terrain In Render** - swap the simulation mesh for the displaced

Material masks

Material Masks and **Auto Masks** give your shader the maps it needs - where sand is fresh, where it is packed, where rock shows through, where it is wet. This is how you drive a believable surface shader without hand-painting.

Things that make shading more interesting

different.

into, with **Layer Thickness**, **Bed Dip**, **Bed Waviness** and **Band Contrast**. This is what makes an eroded dune face look geological.

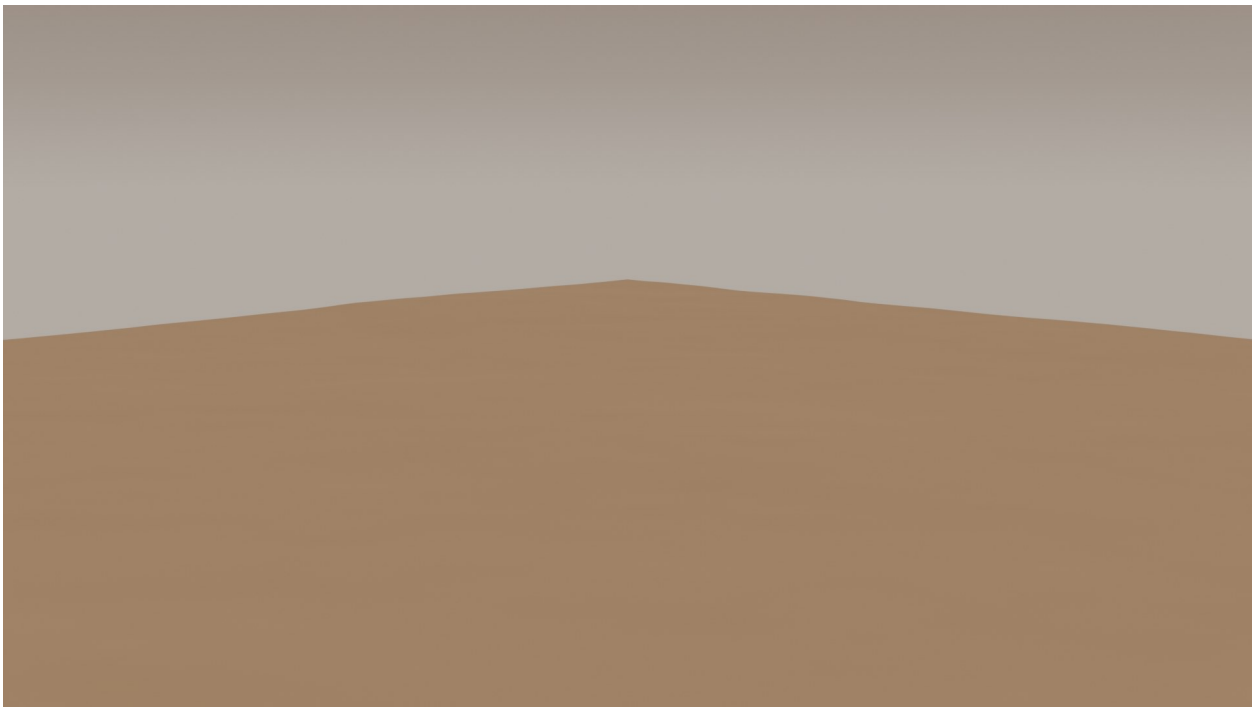
- **Track Sand Age** - how long sand has been sitting. Old and new sand can look
- **Strata / Cross-Bedding** - internal layering exposed when a dune is cut
- **Pebble Density / Grass Density** - scatter points for your own assets.

Field maps

The **Field Maps** panel loads and saves the solver's own fields as images - heights, masks, sand sources, wind gain. Use it to round-trip with other tools, or to reuse a hand-painted mask across several shots.

Tutorial 15 - Making it fast enough

Goal: interactive iteration. **Time:** 10 minutes.



Mars: the same solver under a different planet's gravity and atmosphere

The main lever is Resolution. Cost scales with grid size. Work coarse, raise it at the end. This one decision dominates everything below.

Then, in the Performance panel:

Control	What it trades
Fast 32-bit Solver	Precision for speed. Usually invisible in the result
Display Resolution	Viewport mesh density - cheaper redraws without touching the physics
Viewport Update Every N	Redraw every Nth step instead of every step
Preview Physics	A cheaper model while you are looking around
Proxy Simulation	A coarse stand-in for interaction
Auto Substep / Max Substeps	How finely each step is divided. Fewer is faster, less stable
Avalanche Quality	Exact versus approximate avalanching
Fused GPU Avalanche / Fused GPU Wind Scan	Combined GPU kernels - leave on
Pipelined Playback	Playback one frame behind, for smoother scrubbing
Give The CPU Back	Stops a core spinning while the GPU works. Leave on unless benchmarking

Remember Run Steps. Batching 100 steps with no redraw is much faster than watching 100 frames play.

On the CPU path (no CUDA card): everything above matters more. Start lower, use Run Steps rather than playback, and raise resolution only for the final pass.

Tutorial 16 - Tiles that match their neighbours

Goal: a simulated patch that blends into surrounding terrain. **Time:** 10 minutes.

A simulation domain has edges. If you drop the result into a larger landscape, those edges show. Two tools fix this:

Edge Flat pins a border band, measured in **grid cells**, to the starting terrain. Nothing evolves there, so the edge stays exactly as it began and matches whatever is next to it.

Lower Edges (the edge falloff operator) blends the terrain down at its boundary, with **Edge Width**, **Edge Shape**, **Edge Curve** and **Edge Level**. Setting **Edge Level** to a fixed height and using the *same value on every tile* is what makes separate tiles meet cleanly.

Use them together: falloff for the soft transition, Edge Flat to hold the outermost band still.

Also relevant: boundary mode. Periodic edges wrap sand around from one side to the other - useful for an endless field, wrong for a bounded set. Open boundaries let material leave, with **Input Flux** controlling what drifts in.

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7. The 66 demo scenes

Presets > Demo Scenes > Create Demo Scene. Each builds a complete working scene - terrain, lights, camera, materials - with the settings that make its subject visible.

Use them as documentation you can run. Every significant feature has at least one scene built specifically to demonstrate it, and each scene's settings were chosen because they meet that feature's conditions.

How to learn from one:

1. Build it and look at it.

2. Read the card in the corner - it says what to look at and usually suggests

something to try.

3. Switch the **Interface** to Standard or Advanced and see what the scene

turned on. Every setting is yours to copy.

4. Remove Demo Scene takes it back out and leaves your own objects alone.

The families:

to see what the add-on carries at once.

side in one frame.

rain, frost, vegetation.

- **Grand Scenes** (4, at 400 m) - many systems in one landscape. The best place
- **Showcase scenes** - catalogues that put several related features side by
- **Terrain scenes** - start from a landscape type rather than flat ground.
- **Phenomenon scenes** - isolate one thing: avalanches, sluices, ice, thaws,

Grand Scenes (400 m, many systems at once)

Scene	What it demonstrates
Grand Scene: Desert Expedition	A 400 m landscape with dunes, grains, rock shelters, sources, tracks and airborne dust
Grand Scene: Living Oasis	A 400 m rain-fed basin combining wet shores, plant species, growth and surface weathering
Grand Scene: Winter Storm	A 400 m snowfield with layered drifts, density, slabs, shelter and snow-fed vortices
Grand Scene: Dynamic Coast	A 400 m coast with Flow 2D wind, shallow water, waves, sediment, rain and vegetation

Showcase scenes (several related features in one frame)

Scene	What it demonstrates
Showcase: Object Masks	Three object slots doing three different jobs in one scene: a source box feeding fresh sand, a wind-gain box acting as a gust lane, and rock obstacles casting wind shadows. Every one of these masks can also be painted with the Interactive Brush

Scene	What it demonstrates
Showcase: What Objects Tell The Sand	Three slabs on one moving sand sheet, each holding the ground to a different rule, with every number below measured on the scene itself. BEDROCK is a floor the sand cannot sink below; this one sits above the sheet, so it stands as a rock sill the sand flows around, 1.43 m proud of the field. ERODE is the quiet one: the ground there stops moving altogether, roughness 0.0025 against 0.30 for the sheet beside it, so dunes travel past a patch that never changes. NO-DEPOSIT is the one people mix up with it: nothing new lands, while whatever sand is there is carried off as usual, so the footprint SINKS -- 0.12 m after the warm-up and 0.38 m after 300 steps more. Erode holding still while No-Deposit digs itself out is the whole point
Showcase: Grain Systems	The grain-scale models together: Multi-Fraction winnowing fines out into an armouring coarse-lag pavement, Strata painting cross-bedding on cut faces, Sand Age reading fresh on scoured slip faces, and the physical ripple solver at megaripple scale. Ships with the Surface Shader so all of it is visible
Showcase: Surface Detail	The layer that lives ABOVE the dune, shot from close enough to see it -- from the usual demo distance it is smaller than a pixel. The Micro-Detail pass has been run once already, freezing the dunes as bedrock and running transport again at a finer scale over them, with Crispness, Crests and Slump Allowance set; those three do nothing until the pass is RUN, so press the button again after changing them. De-facet Crests takes out the straight grid zigzag that a one-direction wind leaves along the crest lines, Edge Rounding takes the corner off the brink but not off the flats, and Detail Feeds Sim hands a little of the detail back to the solver so the ripples are sand rather than a decal the wind ignores
Showcase: Surface Shader	How the surface texture is PROJECTED, rather than what it is. A flat top-down projection stretches the grain down every slope and a dune is mostly slope, so this scene switches the scanned sand and the bedrock onto a triplanar projection and turns up the four map-side controls: the scan follows the wind heading and the local downslope, the procedural/scan balance becomes a map instead of one number, and the terrain sits in its own ambient shade. Note the trade it has to make: triplanar and Hex-Tiling cannot both run on the sand, so the scene uses Texture Bombing -- put Hex back and the triplanar tick silently stops doing anything
Showcase: Airborne + Dressing	Everything that is not the heightfield: swept sand-haze particles emitted off exposed crests and settling in the lee, wandering dust devils, a curve imprinted as a rut with berms, and pebble scatter driven by the simulation masks

Terrain scenes (start from a landscape type)

Scene	What it demonstrates
Terrain: Mountain Range	Ridged fractal peaks with sand collecting in the valleys -- the classic case where the rock is the scenery and the dunes are what moves through it
Terrain: Canyon	A terraced, deeply-cut canyon frozen as bedrock with sand fed along the floor, so drifts snake between the walls and pile against them
Terrain: Cellular Mesas	Turbulence terracing gives flat-topped mesas with steep sides; sand streams around them and builds lee tails behind each
Terrain: Ready Dune Field	Wind-aligned ridges as the starting field, so a mature dune sea exists from frame 0 instead of forming over hundreds of steps
Terrain: Beach + Buildings	A shallow shore with a water table, a row of buildings behind the dune line, and sand drifting up against them
Terrain: Buried Town	A block of buildings on an open plain being swallowed by sand: each one casts a wind shadow and grows its own lee drift
Terrain: Crater Basin	A steep-walled basin that traps everything blown into it -- a sand trap, and a good stress test for avalanching on rock
Terrain: Rolling Hills	Soft FBM hills with a full sand cover: gentle relief where the dunes and the ground shape each other
Terrain: Stepped Plateau	Hard terraces frozen as bedrock, with sand only on the steps -- shows the non-erodible rock path and how sand cascades between levels

Scene	What it demonstrates
Terrain: Islands + Water	Scattered high ground in a shallow sea: wet locked shores, dry migrating crests, and rain to tie the two together

Snow, frost and avalanches

Scene	What it demonstrates
Snow: Drifts + Fence	Wind-packed snow drifting behind a slatted fence, with a snow material and cool daylight. Shows the snow solver and obstacle wind shadows building lee tails
Snow: Slab Avalanche	A weak layer buried under a wind slab lets go on a 38 degree face and the debris RUNS OUT onto the flat below. This is the one scene where Slab Dynamics is on: without it the released mass is only ROUTED downhill, which moves it DOWN THE SLOPE rather than out across the flat. MEASURED on this scene, the two models differ by 10.7 % of the released mass: turning Dynamics on shifts 266 units off the upper face and lays them on the runout, concentrated around 75-87 % of the way across. Both models DO reach the valley floor and both stop at the same 95th-percentile distance -- the difference is WHERE the debris piles up, not how far it gets, and in a wide shot it is only 0.12 % of pixels. Look at the runout zone, not at the whole slope. MEASURED, and it is why the scene sculpts its own mountainside rather than using noise: on the stock demo terrain the 95th-percentile slope is 0.103 against a release threshold of tan(34) = 0.675 -- six times too flat, and 220 steps released nothing at all. Try Slab Dynamics 1 -> 0 and watch the pile move UP the runout toward the foot of the face
Snow: Avalanche That Scours Its Path	The same 38 degree face and the same storm as Snow: Slab Avalanche, with Entrainment ON (0.007 m per m of path) and Turbulent Friction set to 400. A real avalanche does not carry only the slab that let go: it scrapes the snow it runs over and grows. MEASURED on this scene, over the same storm: the flow picked up 35 033 units of snow off its path against 9 137 released by the slabs themselves -- 3.8 times the slab -- and 54 342 with the turbulent term on top, because a slower flow lives longer inside its step budget and scrapes further. What you see is the SCOURED TRACKS down the face and a fuller pile at the foot: 5.0 % of the pixels change against Slab Avalanche at the 1 % level, where the two release models of that scene differ by 1.7 %. The turbulent term is here for the SPEED the entrainment law reads, not for the runout: on this 90 m flat it does NOT move the deposit -- 0.432 of the debris ends on the flat at xi 400, 0.432 at 0 and 0.437 at 10, MEASURED -- while the mass-weighted speed drops from 70 to 54 m/s. Try Entrainment 0.007 -> 0 and watch the tracks on the face fill back in
Snow: Avalanche Into A Valley	The same 38 degree face and the same storm as Snow: Slab Avalanche, but the flat is SHORT (30 m) and a 22 degree counter-slope rises behind it -- and all four momentum controls (Carry Momentum, Momentum Rides Along, Sharper Transport, Momentum Crosses Lanes) are ON. Momentum is what lets debris CLIMB the far side; with the four at 0 the flow is hydrostatic and stops at the foot. MEASURED on this valley (512 cells, a 0.5 m slab off the face, friction 0.12): with the four on the front reaches the counter-slope after about 7 s, climbs to 174 m along it (22 m up) by about 11 s, then part slides back and 11 % of the debris settles on the counter-slope; with the four off the flow stops on the flat after about 10 s and never touches it. TIME IS THE TRICK: at 1500 substeps the run is cut after about 3 s and both variants deposit mid-flight on the flat with no run-up at all, so this scene runs 12000 substeps with Slab Flow Rest Speed 0.05 m/s. Try Carry Momentum 1 -> 0: the other three go dead with it and the debris stays on the flat
Sand: Slide Under Gravity	No wind. A rock face sculpted to 45 degrees, steeper than the Repose Angle (34), with a 0.6 m strip of sand and two thicker patches painted on it. The repose relaxation runs every step whether or not the wind blows, so the sand slides on its own and stops where its surface reaches 34 degrees. MEASURED on this scene (512 cells): 91 % on the face when the demo opens, 73 % after 20 steps and at rest from 40 steps on at 69.5 %, with 30.5 % at the foot as talus, mass kept to the last digit; with Avalanche Passes 0 nothing moves at all. Try Repose Angle 34 -> 25 (rest at 49 %) or 40 (85 % and still creeping)

Scene	What it demonstrates
Snow: Sluffs Off A Rock Face	The same rock face at 50 degrees, no wind, and a storm falling straight down. Snow landing above the repose limit sluffs to the foot as it falls; with Avalanche Trigger 6 it holds a few degrees past repose and lets go in events. MEASURED: 30-40 % of what fell on the face is still there when the storm ends, the drops come in steps of up to 0.18 points per step against 0.04 with continuous relaxation. Snow Repose Offset does NOT change this -- it belongs to the slab model (measured 0 / +12 / +20 identical). Try Avalanche Trigger 6 -> 0
Snow: Slab Avalanches Held Back By Gates	The Slab Avalanche face with three plates standing across it at 30, 40 and 50 % of the way down; the storm keeps falling while you Play, slabs keep letting go and the debris is ROUTED downhill (Slab Runs Out 0 on purpose: only the routing sees obstacles). The plates are KEYFRAMED to lift at frames 30, 70 and 110; each lift hands what it held to the next gate. MEASURED on a fixture: without a plate 89 % of the debris ends below its line, with it 45.5 % and the upper half stays put; with Slab Runs Out 1 the plate is ignored (85 % passes). Loose snow would not do: a uniform layer does not slide in the local repose rule, it only retreats from the top edge
Snow: A Thaw That Leaves One Patch	The drift field and the same fence, run through a THAW. Snow lies at Initialize (Initial Snow Depth 0.5), the winter builds the drifts, and then the air goes to +8 C with Melt Rate on -- and the field loses ALL of it. MEASURED over 30 steps: mass of the fresh layer 20 584 with no thaw against 0 with it, cover 98.2 % against 0.0 %. One patch survives, because a snow EMITTER keeps feeding it: Snow Source Rate 0.05 over a box, feathered by Snow Source Smoothing, holds 2 842 of mass and 4.9 % cover when everything around it is bare. MEASURED and the reason this scene keeps the DunesForge surface shader instead of the flat snow material the other snow demos wear: on flat white paint the same thaw changed frame brightness from 0.7232 to 0.7200 and the renders looked identically white -- only the drift relief moved. With the snow mask driving the surface, the thaw repaints 57.1 % of the pixels and the emitter patch another 4.4 %. Try Melt Rate 0.01 -> 0 with the air still warm: melting is Melt Rate PLUS the temperature term, so the warm air alone still takes it, just more slowly. Try Snow Source Rate 0.05 -> 0: the last white patch goes with it
Snow: Fresh, Packed and Drifted	The same fence and the same wind as Snow: Drifts + Fence, with Snow Density switched ON -- so the pack stops being a depth and starts being a MASS. A depth-only solver cannot tell 10 cm of powder from 10 cm of settled snow; MEASURED on the gate fixture those are 10.00 and 50.00 kg/m2, five times apart at the same depth, and across this field the mass runs 0.37 to 17.42 kg/m2. Drifted snow is denser than the snow that fell, because the grains shatter in saltation and pack tighter, so the drift behind the fence settles SHALLOWER and HARDER than the fresh fall around it -- measured mean hardness 0.1204 at Drifted Density 350 against 0.8125 at 100. Try Drifted Snow Density 350 -> 100: the mechanism switches itself off bit for bit and the pack deepens again
Frost: Cold Dawn	Surface frost whitening the damp hollows while the dry crests stay bare sand
Frost: Hoar on Snow	A cold, sheltered snowfield growing surface hoar -- the weak layer a later slab will release over

Water, rain and ice

Scene	What it demonstrates
Oasis: Water + Valley Masks	Sand placed in valleys around a shallow water table with a translucent water plane: wet shores lock in place while dry sand keeps migrating around them
Water: Pools That Flow	The SAME oasis as above, with the water given a depth field instead of one global level -- so every hollow settles to its own height, water runs downhill between them and the shore is wherever the ground says. Built as a direct A/B against the Oasis demo: same terrain, same warmup, one switch different
Water: Wind On The Lake	The flowing field with a 30 m/s wind gripping the surface. The water piles up on the DOWNWIND shore at a slope of exactly tau/(rho g h), so the shoreline moves and the pools lean. MEASURED on this terrain at 30 m/s against no wind, after 3 s of water time: 7699 cells cross between wet and dry -- 4010 flooded and 3689 drained -- and the surface stands 13.7 mm apart at most. The wind here is the ANALYTIC one, so it is the same everywhere; switch Wind Model to Flow 2D and give the flow an Air Layer to have crests and lee pockets push the water differently

Scene	What it demonstrates
Water: Reservoir Behind A Sluice Gate	A rock plateau with a 6 m basin, a dam with a 16 m gap and a valley below. The basin holds 1.5 m of water above the sill; a red gate in the gap is an ordinary obstacle that, with Water Sees the Rocks on, is part of the water bed and holds the water back. It is KEYFRAMED to lift 24 m from frame 20: press Play and the reservoir pours down the valley as a shallow-water flood. MEASURED on a fixture of this geometry: without the gate 7.8 % of the water is below the dam after 20 s and 34.8 % after 120 s, with the gate 0.0 % for 120 s, gate removed at 40 s -> the same curve shifted by 40 s; mass 1.0000. Try Water Sees the Rocks off: the flood starts at once
Ice: Pack, Ridges and Leads	The ice mechanics of 1.3.134-1.3.138 in one scene: a frozen basin kept ALIVE by a steady inflow, so the pack never settles. The current carries the sheet, piles it against the shore into RIDGES where it converges and pulls it open into LEADS where it diverges -- all three out of one conservative flux, with no extra rule. MEASURED, and this is why the demo exists: on the Pools That Flow terrain the water reaches equilibrium and the ice sampled a LIVE current in only 2 of 20 steps, so none of it could be seen. With the inflow on it is 20 of 20. Try Ice Yield Strength 20 -> 0 and watch the ridge turn into an even thickening
Rain: Wet Sand + Plants	Storm cells sweep a formed dune field: the dunes are built DRY first, then the rain arrives, so you can see what it changes. Damp patches go dark and matte and stop moving, the lee of the rocks stays dry in the rain shadow, and vegetation greens up where the root zone holds water and then pins the sand down
Rain: Monsoon Shutdown	The heavy end of the rain model: near-total coverage, slow cells and strong infiltration, so almost the whole field goes wet and almost all of it stops moving. Shows what OUTLIVES the rain -- a drying crust plus the vegetation it fed, which keep holding the sand after the weather has gone
Rain: Wet Oasis	Water table AND rain at the same time: two independent sources feeding one wetness mask, each with its own hold strength. Plants ring the standing water where the root zone never dries out. Good for checking that the two systems combine rather than fight
Rain: After The Storm	The field is soaked and then the rain STOPS. Wind Drying makes exposed crests dry first while sheltered lee slopes stay damp, so the dunes wake up unevenly instead of all at once -- the clearest demonstration of the wind/moisture coupling
Rain: Drops On Water	A close-up of the rain IMPACT layer, framed on a shoreline so you can see both behaviours at once: rings spreading where a drop lands in standing water, and splash craters with dark wet spots where it lands on the shore. The camera sits low and near on purpose -- an impact is about 5 cm across, so from the usual demo distance every one of them would be a single pixel
Rain on Snow: Crust	Warm rain falling on fresh snow: it takes depth off the pack and leaves what is left bonded into a crust
Rain on Snow: Loaded Slab	The same crust with fresh snow dumped on top of it -- the layered setup a slab releases over

Vegetation scenes

Scene	What it demonstrates
Plants: Two Species Side By Side	A succulent and a grass on the same dune, which is the commonest arrangement on a real one. Drought Tolerance used to be ONE number for the whole scene, so the two could not stand together; above 0 Second Species turns it into a FIELD of patches, deterministic in Seed. MEASURED on the gate fixture, cover of the tolerant plant against the sensitive one: 0.4973 / 0.7078 with one species (a +0.2105 gap that is just the two thresholds) against 0.5087 / 0.7811 with the species map on -- the gap widens to +0.2725 because each plant now sits where it can. The two colours are what makes it readable: Vegetation Colour is the grass, Vegetation Colour B the succulent. Try setting Tolerance (Species B) equal to Drought Tolerance: the map stops mattering and the result goes back to being identical to the bit
Vegetation: Nabkha Hummocks	Scattered plant clumps on a sand-fed plain, each one trapping a mound with a downwind tail -- the nabkha (coppice dune) landform. Built with the patch-shape controls open, so the clumps are ragged, come in a range of sizes and grow in stands instead of being identical circles on a uniform scatter, and with the clumps taking their own darker colour so a hummock reads as a bush and not as a green stain

Scene	What it demonstrates
Vegetation: Rain-Fed (nothing grows on its own)	The only scene where NOTHING grows by itself: Veg Growth is set to zero, so every plant you see was put there by rain. Growth From Rain is a multiplier and cannot help at zero growth, so Rain Seeds Growth -- which adds its own rate from the root zone -- is the only term that can raise a single stem, which makes the scene its own control. Spread sits at 1.0 so growth happens only where plants already are: clumps thicken and creep at their edges while two thirds of the ground stays bare, and that bare sand is what lets you see the second half of the loop, Plants Shade The Ground, holding moisture under the cover that the rain grew
Plants: Valleys Only (Slope and Height Windows)	The blowout field again, with the vegetation kept OUT of the top 40 % of the relief by Height Max 0.6, Limits Apply to Seeding ON and Edge Noise 0.5. The window is what keeps a real field from greening over: plants hold the hollows and the crests stay bare sand. MEASURED on this scene after the 150 step warm-up, cover on the top third of the relief: 0.992 with no window against 0.040 with it, and 0.000 at frame 0 because Limits Apply to Seeding never lays it there; without that switch seeding covers the crests first and burial has to clear them. The relief is 8.0 here, ten times the Blowouts scene, MEASURED: at 0.8 the dunes rebuild the height field during the warm-up, the crests at the end are not the crests the seeding obeyed, and the window keeps only 0.676 of them bare (2.0 -> 0.292, 4.0 -> 0.098, 8.0 -> 0.040). Burial is 0.5, not 2.0, because at 2.0 the crests are bare with or without a window and there would be nothing to show. MEASURED and fixed in this version: Edge Noise used to jitter the OPEN limits too, so a height-only window refused 45.3 % of a perfectly flat floor; it now frays only the limit you set. Try Invert Limits: the crests green and the hollows go bare. Try Edge Noise 0.5 -> 0 and the tree-line turns into a contour
Vegetation: Blowout Hollows	The opposite starting point: a fully vegetated coastal grassland with bare gaps punched in it. The wind scours the unprotected gaps into deflation hollows while the anchored surroundings hold, which is how a blowout -- and eventually a parabolic dune -- gets going
Vegetation: Green Retreats	A field that starts GREEN EVERYWHERE and loses it: drought dieback takes the cover off the dry ridges while the plants ringing the water hold, so the green pulls back to the shoreline instead of simply fading. Built the way round it is for a measured reason -- with the usual near-bare start the same physics moves 41% of the dry-zone cover and changes 1% of the rendered pixels, because 0.068 and 0.040 both read as bare sand. Starting green makes the loss an AREA that shrinks. MEASURED on this scene after 90 steps against the same scene with Drought Dieback at 0, counting only VISIBLE LAND (ground above the waterline -- cover under an opaque water surface is not something you can see): the strongly green share of the land falls from 78.4% to 11.0%, and mean cover on land from 0.549 to 0.165. The pool itself is half a percent of the frame while the wetness reaches 77% of it, so the green ring sits on LAND rather than under water. Turn Drought Dieback to 0 and the green stays put

Flow scenes (wind that goes around things)

Scene	What it demonstrates
Water: Ripples Carried By The Current	The Pools That Flow basin with a DEAD SWELL pumping it, so the current never settles -- and that is the whole condition of this scene. Flow Ripples makes the fine chop read the FLOW FIELD instead of an independent noise: the pattern stretches along the current and twists where it turns. MEASURED on this scene: straight after the warm-up the basin is in EQUILIBRIUM -- peak speed 0.0 and 0.0 % of cells moving -- so the knob changes 0.000 % of the pixels, because there is no flow to read. With the swell on: peak 36.15, mean 0.218, 70.4 % of cells alive, and the same knob changes 4.69 % of the pixels on the built scene (4.84 % on the gate fixture) against 0.000 % render noise. Try Flow Ripples 2.5 -> 0: the chop goes back to an independent noise that does not know where the water is going. Try Swell Height 0: the basin settles, the current dies, and the knob stops mattering -- which is the same trap the Ice: Pack scene documents
Flow 2D: Canyon Jet	Two rock walls with a gap. The solved wind speeds up through the gap because the same air has to pass a narrower opening -- a result of mass conservation, not of a correction term
Flow 2D: Recirculating Wake	A single bluff rock. Behind it the flow REVERSES and turns back on itself, which the analytic wind cannot do at all -- it has a lee shadow but no momentum

Scene	What it demonstrates
Flow 2D: Wind Off a Wall	An angled barrier that turns the wind sideways. Tracers and the direction overlay are on, so the bounce is visible as motion and as colour

Other phenomena

Scene	What it demonstrates
Desert: Dunes + Obstacles	Classic sand dune field with rock obstacles casting wind shadows, pebble scatter, camera, sun and a sand material. Pre-runs the simulation so dunes are already formed
Another Planet: Mars	The same ORGANIC dune field as the other scenes with ONE switch different: Other Planet, set to Mars. Martian air is sixty times thinner than ours (0.020 against 1.20 kg/m3) and the saturation length divides by air density, so the same physics builds dunes far larger on the same ground. Turn Other Planet off and the scene rebuilds itself at Earth scale, which is the whole point of the comparison; scale the terrain object up if the Martian dunes outgrow the default 200 m domain
Rock + Sand Source	Start Terrain = Rock workflow: a rocky FBM terrain frozen as bedrock, with sand fed only from an upwind source zone so drifts grow from nothing over bare rock
Phenomena: Playa Crust	A dried lake bed: the crust sets and cracks into plates, and the wind deflates along the cracks while the plates stand
Phenomena: Wind Erases Tracks	A rut imprinted across the field, fading faster where the wind is strongest and surviving in the lee
Temperature: Cold Front	One cloud over a warming scene: the same storm lays snow while the air is below zero and wets the ground once it is above
Temperature: Spring Thaw	A snowfield under a rising sun -- melt on degree-days, so it stays put until the air crosses zero and then goes quickly
Freeze: Locked Shoreline	Wet sand along a water table, frozen solid: the damp band stops eroding entirely while the dry dunes behind it keep migrating
Freeze: Thawed Shoreline	The same shoreline above freezing -- the control that shows the lock is the ice and not the water
Phenomena: Snow Fence	A POROUS blocker: the long downwind drift a real snow fence builds, next to the short steep pile a solid wall makes
Phenomena: Snow Layer Stack	A season of storms as dated layers, with a buried weak layer that decides where a slab releases
Phenomena: Sublimation + Hoar	Snow leaving as vapour on the exposed crests while surface hoar grows in the sheltered hollows
Phenomena: Tidal Coast	A moving shoreline: the damp band sweeps up and down the beach and the dunes behind it advance in steps

8. Glossary

Terms that appear in the interface and in this documentation.

Term	What it means
Aeolian	Shaped by wind. The whole field this solver models
Angle of repose	Steepest slope loose grains hold before avalanching (~34 deg dry sand)
Armouring / lag	Coarse grains left behind by wind, protecting finer material beneath
Avalanche	Relaxation of an over-steep face back to the repose angle
Barchan	Crescent dune with horns pointing downwind; forms where sand is scarce
Bedrock	Ground that does not erode. Sand collects on it

Term	What it means
Blowout	A hollow opened by wind where vegetation has failed
Crest	The top line of a dune, where windward and lee faces meet
Deflation	Removal of material by wind - the opposite of deposition
Dune Scale (Ls)	The solver's own dune length scale, mapped onto your world
Flux	The amount of material moving past a point
Height field	The grid of heights the solver actually simulates
Lee side	The sheltered, downwind side. Where sand drops
Megaripple	Large ripple with coarse-grained crests; needs two grain sizes
Nabkha	A mound of sand trapped and held by a plant
Pavement	Surface armoured with coarse grains after fines are blown away
Repose	See angle of repose
Saltation	Grains bouncing along the surface - the main transport mechanism
Separation	Wind detaching from the surface behind a crest, leaving a sheltered pocket
Shear / shear stress	The wind's grip on the surface. What actually moves sand
Sintering	Snow hardening with age. Why old drifts persist
Slab	A cohesive snow layer that can release as a unit
Star dune	Dune with several arms, formed by winds from multiple directions
Supply	How much loose material is available
Transverse ridge	Long ridge across the wind, formed where sand is plentiful
Windward	The side facing the wind. Gets scoured
Winnowing	Wind selectively removing fine grains

9. When something looks wrong

Nothing is happening

Check	Fix
Is **Wind Strength** zero?	Set it to `1.0`
Is there loose sand?	Raise **Initial Sand Depth** or **Sand Supply**
Did you press **Initialize Simulation**?	Nothing runs before initialisation
Are the dunes **frozen**?	Check **Freeze Dunes** in the main panel
Did you keyframe it?	Keyframes are read by **Play** only - see §3.3

It does not look right

Symptom	Likely cause
Dunes are the wrong size	Change **Dune Scale**, not the terrain dimensions
Everything is a smooth blob	Sand supply too high, no relief, or not enough steps yet
Dunes look like noise, not dunes	Not enough steps. Dune fields take time to organise
Surface is faceted	Raise **Render Subdivision**, or look at the de-facet controls
Crests look chopped off	Check **Crest Cap** - it limits crest shear on purpose
A combined start looks empty	Placement filters are ANDed; try **Combine with OR** or widen the bands

Symptom	Likely cause
Edges of my tile do not match	See Tutorial 16
Sand wraps around to the other side	Boundary mode is periodic. Use an open boundary

It is too slow

Read Tutorial 15. In short: lower **Resolution**, use **Run Steps** instead of playback, and check whether the GPU backend is actually active (**GPU Self-Check** in the main panel).

Something changed that I did not change

leak into each other.

that.

- **Applying a preset resets shared controls** - deliberately, so presets cannot
- **Some presets also replace the terrain.** Those say so in their names.
- **Switching Interface level changes nothing.** If a value moved, it was not

I need to see what the solver thinks

Debug / State reports the solver's current state in numbers. The **Diagnostics** button opens the folder holding the diagnostic log, which records what the solver actually did - that is the file to send when asking for help.

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10. Checklists

Before you start a shot

- [ ] Decided sand-rich or sand-starved (**Sand Supply**, **Initial Sand Depth**)
- [ ] Terrain source chosen (procedural, object, or heightmap)
- [ ] **Start Terrain = Rock** set correctly for what you want
- [ ] **Resolution** set low for iteration
- [ ] **Seed** noted, if you want to reproduce this exact field

Before you call the simulation finished

- [ ] Run long enough that the field has organised, not just roughened
- [ ] **Dune Scale** checked against your camera framing
- [ ] Obstacles interacting the way you expect
- [ ] Edges handled, if this is a tile (Tutorial 16)
- [ ] Looked at it from the render camera, not just the working view

Before you hand it off

- [ ] Mesh committed, or maps baked (Tutorial 14)
- [ ] Material masks generated if the shader needs them
- [ ] Demo-scene leftovers removed (**Remove Demo Scene**) if you started from one
- [ ] Simulation stopped costing you time you are no longer using

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Where to go next

REFERENCE.md - every control, panel by panel, with its default and what it is for. Use it to look things up, not to read front to back.

Useful jumping-in points:

- Wind - direction, strength, variation, wind models
- Physics - repose angle, avalanching, transport
- Objects / Masks - obstacles, sources, tracks
- Water / Moisture - water table, flow, rain
- Vegetation - growth, burial, species
- Surface Shader - how the surface is textured
- Surface Detail - ripples and micro-relief
- Export / Bake - getting maps and meshes out
- Performance - the speed/quality trade-off